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# Effects of dietary mulberry, Korean red ginseng, and banaba on glucose homeostasis in relation to PPAR-α, PPAR-γ, and LPL mRNA expressions

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**Abstract**

Despite lack of scientific evidences to support its therapeutic efficacy, the use of herbal supplements has significantly increased. The purpose of this study was to evaluate the effects of traditional anti-diabetic herbs on the progress of diabetes in db/db mice, a typical non-insulin-dependent model. Five different experimental diets were as follows: control diet, 0.5% mulberry leaf water extract diet, 0.5% Korean red ginseng diet, 0.5% banaba leaf water extract diet, and 0.5% combination diet (mulberry leaf water extract/Korean red ginseng/banaba leaf water extract, 1:1:1). Blood levels of glucose, insulin, HbA1c, and triglyceride were measured every 2 weeks. At 12 weeks of age, animals were sacrificed, and tissue mRNA levels of PPAR-α, PPAR-γ, and LPL were determined. Results indicated that mulberry leaf water extract, Korean red ginseng, banaba leaf water extract, and the combination of above herbs effectively reduced blood glucose, insulin, TG, and percent HbA1c in study animals ($p<0.05$). We also observed that the increased expressions of liver PPAR-α mRNA and adipose tissue PPAR-γ mRNA in animals fed diets supplemented with test herbs. The expression of liver LPL mRNA was also increased with experimental diets containing herbs. The efficacy was highest in animals fed the combination diet for all of the markers used. These results suggest that mulberry leaf water extract, Korean red ginseng, banaba leaf water extract, and the combination of these herbs fed at the level of 0.5% of

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the diet significantly increase insulin sensitivity, and improve hyperglycemia possibly through regulating PPAR-mediated lipid metabolism.


*Keywords:* C57BL/KsJ-db/db mice; Mulberry; Korean red ginseng; Banaba; PPAR-α; PPAR-γ; LPL

## Introduction

Herbal remedies have been used in medical practices for many years in East Asian countries and account for approximately 80% of medical treatments in the developing countries (Gesler, 1992). The use of herbal supplements in western parts of the world has steadily increased over the last decade. A report from United State showed that the use of herbal supplements increased 380% between 1990 and 1997 (Eisenerg et al., 1998). However, limited scientific evidences are available for the efficacy of herbal medicine. Also, the precise mechanisms of actions by active ingredient(s) are rarely understood.

Despite numerous preventive strategies and medical treatments, 300 million people are expected to develop non-insulin-dependent diabetes mellitus (NIDDM) in 2025 worldwide (Scheen, 2000; Seidell, 2000). As a result, a growing number of people are turning to alternative therapies including herbal medicines. Ginseng has been used as a traditional anti-diabetic remedy for many years in countries such as Korea and China. The efficacy of ginseng has also been reported in a number of animal studies, although many of them used insulin-dependent diabetes mellitus (IDDM) models. Several studies have suggested that ginseng intake possibly control postprandial hyperglycemia both in healthy and diabetic individuals (Vuksan et al., 2000; Soonthornpun et al., 1999; Turner, 1998); however, their mechanisms of actions are not clearly understood. Mulberry root bark or leaf extracts were shown to possess hypoglycemic effects in IDDM animal models (Hikino et al., 1985; Chen et al., 1985). 2-*O*-α-D-galactopyranosyl-DNJ (GAL-DNJ) and fagomine have been suggested as potent antihyperglycemic compounds in mulberry roots and leaves (Chen et al., 1995). Banaba leaf extract is also shown to decrease blood glucose in genetically diabetic (KK-AY) mice (Carew and Chin, 1961), possibly by increasing the uptake of glucose by adipocytes (Emmerson, 1996). Corosolic acid was identified as an active ingredient in banaba extract (Harris et al., 1999).

Although the causes of NIDDM are not fully understood, central obesity and insulin resistance, which lead to hyperinsulinemia, are known risk factors for diabetes (Lenhard and Gottschalk, 2002). Main features of insulin resistance include dyslipidemia, which is characterized by high triglyceride, low HDL, small and dense LDL (Peter et al., 2003) in the blood. Epidemiological studies have suggested that the risk of cardiovascular disease is two- to six-fold in NIDDM patients (Pan et al., 1986; Stamler et al., 1993), and both insulin resistance and dyslipidemia may contribute to the increased cardiovascular risk in these patients. Therefore, dyslipidemia as well as hyperinsulinemia should be major intervention targets to reduce a risk of developing NIDDM and NIDDM-related CVD risk. In recent studies, PPARs (peroxisome proliferator-activated receptors) agonists are known to alleviate dyslipidemia and insulin resistance, and many synthetic ligands for PPARs including thiazolidinediones have been successfully used as insulin sensitizers in NIDDM patients (Grommes et al., 2004).

Therefore, the purpose of this study was to evaluate the efficacy of popular traditional anti-diabetic herbal supplements, alone or in combination, to retard the progress of the disease using db/db mice as a typical NIDDM model. We also examined if the actions of these herbal remedies are partly mediated through PPARs.

## Materials and methods

### *Animals and diets*

A total of 50 animals (4-week-old C57BL/KsJ-db/db, male; ORIENT Co., Seoul, Korea) were used. The mice were housed (two per cage) in environmentally controlled conditions with a 12-h light/dark cycle. Animals were allowed to acclimatize to the laboratory environment for 1 week and randomly divided into five groups ($n = 10$ group) as given below: group I—control, group II—0.5% mulberry leaf water extract diet, group III—0.5% Korean red ginseng diet, group IV—0.5% banaba water extract diet, group V—0.5% combination (mulberry leaf water extract/Korean red ginseng/banaba leaf water extract, 1:1:1) diet. The compositions of the experimental diets are shown in Table 1. Herb samples were analyzed for carbohydrate, fat and protein, and experimental diets were adjusted for these macronutrients to provide diets with same energy density. Animals were maintained on experimental diets for 12 weeks with free access to food and water.

### *Plant materials*

The mulberry (*Morus alba* L.) leaf water extract powder, Korean red ginseng (*Panax ginseng* C.A. Mayer) powder, and banaba (*Lagerstroemia speciosa* L.) leaf water extract powder were kind gifts from the National Agricultural Cooperative Federation (Jeung-Pyeong, KOREA). In brief, leaves of mulberry and banaba were powered and extracted with 8 and 10 times of hot water (85 °C) for 10 h, respectively. The extracts were filtered, concentrated, and dried under vacuum. The dried samples were powered at 25 °C.

Table 1
Composition of the experimental diets (%)

| Ingredient | Control | Mulberry leaf extract | Korean red ginseng | Banaba leaf extract | Combination |
|---|---|---|---|---|---|
| Casein | 20 | 19.98 | 19.95 | 19.99 | 19.97 |
| Starch | 15 | 14.94 | 14.67 | 14.77 | 14.79 |
| Sucrose | 50 | 49.58 | 49.89 | 49.73 | 49.94 |
| Corn oil | 5 | 4.99 | 4.99 | 4.99 | 4.99 |
| Cellulose | 5 | 5 | 5 | 5 | 5 |
| DL-Methionine | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 |
| Mineral mix[a] | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| Vitamin mix[a] | 1 | 1 | 1 | 1 | 1 |
| Choline chloride | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Mulberry leaf | – | 0.5 | – | – | 0.17 |
| Red ginseng | – | – | 0.5 | – | 0.17 |
| Banaba leaf | – | – | – | 0.5 | 0.17 |
| Total | 100 | 100 | 100 | 100 | 100 |

[a] Mineral and vitamin mixture used had composition of AIN-76 diets.

### *Preparation of the blood and tissue samples*

Blood samples were collected in the fasting state from the retro-orbital sinus once every 2 weeks. To determine insulin, HbA1c (hemoglobin A1c), and triglyceride levels, the plasma was separated, frozen immediately, and stored at $-70$ °C until assayed. After 12 weeks of feeding period, mice were sacrificed by exsanguination of the heart under light ether anesthesia and blood was collected by cardiac puncture in 5% EDTA vials. Adipose tissues and livers were taken and quickly frozen for the analysis of PPAR-α, PPAR-γ, and LPL mRNA expressions.

### *Fasting blood glucose determination*

Fasting blood glucose levels were determined by collecting blood samples drawn from tail vein once every 2 weeks. Approximately 50 μl of a fresh blood sample was placed on each of duplicate test strips and the glucose content was read in a validated One Touch Basic glucose measurement system (Lifescan Inc., Milpitas, CA, USA).

### *Other biological assays*

Plasma insulin was measured using a Insulin ELISA kit (Linco, St. Charles, MO, USA) with rat insulin as a standard. Percent HbA1c was measured with a Hemoglobin A1c kit (BioSystems S.A., Barcelona, Spain). Plasma triglyceride was measured using a TG kit (Green Cross, Yong-In, Korea).

### *Determination of PPAR-α, PPAR-γ, and LPL mRNA expression*

Total RNA was isolated using Trizol reagent (Sigma, St. Louis, MO, USA). An aliquot of 5 μg of total RNA from each sample was reverse-transcribed to cDNA using the First-strand cDNA Synthesis kit (Pharmacia LKB Biotechnology, Uppsala, Sweden). PCR amplication of 1 μl of cDNA was carried out in a final volume of 50 μl with an MJ Research PTC-0150 MiniCycler (MJ Research, Inc., Waltham, USA), using a HotStarTaq (Qiagen Inc., Valencia, CA, USA). PCR oligonucleotide sequences used were as follows: β-actin (5′-primer: GAGCTATGAGCTGCCTGACG, 3′-primer:AGTTTCATGGATGCCA-CAGGA), PPAR-α (5′-primer: CCTCTTCCCAAAGCTCCTTCA, 3′-primer: CGTCGGA-CTCGGTCTTCTTG), PPAR-γ (5′-primer: GGTTGAACAAGAGATGCCATT CT, 3′-primer: AATGCGAGTGGTCTTCCATCA), and LPL (5′-primer: CGCTCCATTCATCTCTTCA, 3′-primer: CTTGTTGATCTCATAGCCCA). Cycling conditions were as follows: 94 °C for 30 s, 50–68 °C for 30

Table 2
Body weight changes of mice fed experimental diets for 12 weeks

| | 0 week | 2 weeks | 4 weeks | 6 weeks | 8 weeks | 10 weeks | 12 weeks |
|---|---|---|---|---|---|---|---|
| Control | 25.74 ± 5.1 | 33.52 ± 5.5 | 38.75 ± 4.7 | 40.19 ± 7.8 | 43.85 ± 5.9 | 47.15 ± 1.8 | 50.33 ± 4.6 |
| Mulberry | 25.67 ± 6.9 | 34.00 ± 4.3 | 36.20 ± 4.9 | 41.20 ± 7.4 | 43.55 ± 4.7 | 46.48 ± 6.7 | 47.21 ± 4.2 |
| Red ginseng | 27.33 ± 1.1 | 34.12 ± 4.4 | 37.91 ± 5.2 | 40.39 ± 5.6 | 42.12 ± 6.7 | 45.12 ± 5.9 | 46.72 ± 6.1 |
| Banaba | 26.16 ± 6.4 | 35.16 ± 2.9 | 38.85 ± 5.2 | 41.09 ± 4.5 | 43.16 ± 4.3 | 44.96 ± 4.1 | 48.97 ± 3.2 |
| Combination | 26.96 ± 7.2 | 34.19 ± 1.9 | 37.95 ± 4.2 | 41.31 ± 2.2 | 43.25 ± 2.9 | 42.96 ± 8.1 | 45.39 ± 2.0 |

Values are mean±S.D. of 10 rats in each group.

Table 3
Fasting plasma glucose levels of mice fed experimental diets for 12 weeks (mg/dl)

| | 0 week | 2 weeks | 4 weeks | 6 weeks | 8 weeks | 10 weeks | 12 weeks |
|---|---|---|---|---|---|---|---|
| Control | $232.9 \pm 56.2^a$ | $252.7 \pm 70.3^a$ | $293.8 \pm 40.2^a$ | $337.9 \pm 44.8^a$ | $391.2 \pm 30.7^a$ | $410.3 \pm 60.6^a$ | $415.1 \pm 30.2^a$ |
| Mulberry | $220.7 \pm 66.1^a$ | $262.8 \pm 44.3^a$ | $292.9 \pm 33.9^a$ | $320.0 \pm 39.8^a$ | $347.9 \pm 18.9^b$ | $395.3 \pm 43.2^a$ | $387.3 \pm 13.1^b$ |
| Red ginseng | $235.7 \pm 31.9^a$ | $256.0 \pm 50.9^a$ | $300.2 \pm 20.5^a$ | $332.2 \pm 46.4^a$ | $347.0 \pm 10.7^b$ | $391.5 \pm 55.1^a$ | $370.9 \pm 18.7^c$ |
| Banaba | $225.8 \pm 30.8^a$ | $276.8 \pm 46.2^a$ | $297.6 \pm 33.7^a$ | $322.0 \pm 55.8^a$ | $344.0 \pm 11.2^b$ | $399.8 \pm 33.7^a$ | $395.1 \pm 16.8^b$ |
| Combination | $216.6 \pm 40.5^a$ | $276.8 \pm 31.7^a$ | $291.2 \pm 40.6^a$ | $310.7 \pm 20.7^a$ | $341.4 \pm 21.1^b$ | $388.2 \pm 40.9^a$ | $364.7 \pm 18.4^c$ |

Values are mean ± S.D. of 10 rats in each group. Means with different superscripts (a, b, c) within a column are significantly different from each other at $p < 0.05$ as determined by Duncan's multiple range test.

s, and 72 °C for 60 s, for 35 cycles after an initial step of 95 °C for 15 min. A final elongation step of 72 °C for 10 min completed the PCR. The products were then analyzed by 2% agarose gel electrophoresis.

### *Statistical analysis*

Values are expressed as means ± S.D. Duncan's multiple range test was performed to determine significant difference among the groups. A $p < 0.05$ was considered as statistically significant.

## Results

### *Body weight*

The body weights of the animals fed either control diet or herb diets were increased steadily (Table 2). Starting at week 10, body weight increases in mice fed herb diets were lower compared to that of the control mice, although no statistical significance was found.

### *Blood glucose level*

Table 3 shows the time-course changes in fasting blood glucose level for 12 weeks. Animals fed diets containing herbs showed significantly decreased levels of fasting blood glucose. At week 12, animals fed mulberry leaf extract, red ginseng, banaba leaf extract, and the combination diet showed reduced levels of fasting blood glucose by 6.7%, 10.8%, 4.8%, and 12.2%, respectively, compared to the control animals ($p < 0.05$).

Table 4
Blood insulin levels of mice fed experimental diets for 12 weeks (ng/ml)

| | 0 week | 2 weeks | 4 weeks | 6 weeks | 8 weeks | 10 weeks | 12 weeks |
|---|---|---|---|---|---|---|---|
| Control | $2.18 \pm 0.11^a$ | $3.16 \pm 0.98^a$ | $3.95 \pm 1.10^a$ | $4.39 \pm 2.01^a$ | $5.50 \pm 0.01^a$ | $7.39 \pm 0.15^a$ | $9.30 \pm 0.55^a$ |
| Mulberry | $2.19 \pm 0.13^a$ | $3.15 \pm 0.21^a$ | $3.99 \pm 2.01^a$ | $4.32 \pm 1.52^a$ | $4.92 \pm 1.10^b$ | $5.52 \pm 0.78^b$ | $6.01 \pm 1.20^b$ |
| Red ginseng | $2.20 \pm 0.02^a$ | $3.13 \pm 1.54^a$ | $3.88 \pm 1.74^a$ | $4.21 \pm 1.17^a$ | $4.50 \pm 0.14^c$ | $5.10 \pm 1.09^b$ | $5.55 \pm 1.33^c$ |
| Banaba | $2.15 \pm 0.31^a$ | $3.20 \pm 1.20^a$ | $3.94 \pm 1.82^a$ | $4.30 \pm 0.19^a$ | $5.10 \pm 0.03^b$ | $5.66 \pm 1.01^b$ | $6.24 \pm 0.97^b$ |
| Combination | $2.17 \pm 0.94^a$ | $3.17 \pm 0.97^a$ | $3.84 \pm 1.55^a$ | $4.19 \pm 1.68^a$ | $4.32 \pm 0.99^c$ | $4.92 \pm 1.55^c$ | $5.07 \pm 0.87^d$ |

Values are mean ± S.D. of 10 rats in each group. Means with different superscripts (a, b, c, d) within a column are significantly different from each other at $p < 0.05$ as determined by Duncan's multiple range test.

Table 5
Blood glycated hemoglobin (HbA1c) levels of mice fed experimental diets for 12 weeks (%)

| | 0 week | 2 weeks | 4 weeks | 6 weeks | 8 weeks | 10 weeks | 12 weeks |
|---|---|---|---|---|---|---|---|
| Control | $5.73 \pm 0.6^{a}$ | $7.06 \pm 1.1^{a}$ | $7.15 \pm 2.1^{a}$ | $8.03 \pm 0.6^{a}$ | $9.31 \pm 01.0^{a}$ | $11.34 \pm 1.7^{a}$ | $12.64 \pm 1.2^{a}$ |
| Mulberry | $5.43 \pm 1.6^{a}$ | $7.03 \pm 3.4^{a}$ | $6.52 \pm 0.7^{a}$ | $6.06 \pm 0.9^{b}$ | $7.95 \pm 0.7^{b}$ | $9.10 \pm 0.2^{b}$ | $10.41 \pm 0.7^{b}$ |
| Red ginseng | $6.00 \pm 1.7^{a}$ | $6.24 \pm 0.8^{b}$ | $5.87 \pm 1.1^{b}$ | $6.02 \pm 0.9^{b}$ | $8.85 \pm 1.5^{b}$ | $8.45 \pm 1.7^{b}$ | $8.99 \pm 0.7^{b}$ |
| Banaba | $5.46 \pm 1.5^{a}$ | $7.62 \pm 2.3^{a}$ | $5.36 \pm 0.5^{b}$ | $6.14 \pm 1.3^{b}$ | $8.25 \pm 1.3^{b}$ | $9.74 \pm 0.4^{b}$ | $10.82 \pm 1.6^{b}$ |
| Combination | $5.28 \pm 0.7^{a}$ | $6.16 \pm 0.5^{b}$ | $5.10 \pm 0.5^{b}$ | $6.03 \pm 0.8^{b}$ | $8.12 \pm 1.0^{b}$ | $8.40 \pm 1.8^{b}$ | $8.55 \pm 2.2^{b}$ |

Values are mean ± S.D. of 10 rats in each group. Means with different superscripts (a, b, c) within a column are significantly different from each other at $p<0.05$ as determined by Duncan's multiple range test.

### *Blood insulin levels*

Blood insulin level was also measured once every 2 weeks. Control animals showed 4-fold increase in insulin level at week 12 compared to the baseline (Table 4). Starting at week 8, animals on herb diets showed significantly lower level of blood insulin compared to the control animals ($p<0.05$). At week 12, mulberry leaf extract, red ginseng, and banaba leaf extract supplementation reduced blood insulin level by 35.3%, 40.3%, 32.9%, and 45%, respectively.

### *Blood glycated hemoglobin (HbA1c) levels*

We evaluated the percent of hemoglobin nonenzymatically glycated (percent HbA1c), as an integrated measure of long-term blood glucose regulation. Animals fed red ginseng diet and combination diet showed significantly decreased level of HbA1c at week 2 compared to the control animals ($p<0.05$) (Table 5). After 6 weeks of experimental diets, the percents HbA1c level was significantly decreased ($p<0.05$) in all animals fed herb diets and the effect of the combination diet was most significant (Table 5).

### *Blood triglycerides*

Blood triglyceride levels were significantly decreased in all of the treatment groups ($p<0.05$) compared to that of the control animals (Table 6). Similar to above results, the combination diet was the most effective in reducing blood TG.

Table 6
Blood triglyceride (TG) levels of mice fed experimental diets for 12 weeks

| | Triglyceride (mg/ml) |
|---|---|
| Control | $150.4 \pm 3.68^{a}$ |
| Mulberry | $125.0 \pm 1.36^{c}$ |
| Red ginseng | $124.4 \pm 7.67^{c}$ |
| Banaba | $140.6 \pm 2.56^{b}$ |
| Combination | $104.6 \pm 10.64^{d}$ |

Values are mean ± S.D. of 10 rats in each group. Means with different superscripts (a, b, c, d) within a column are significantly different from each other at $p<0.05$ as determined by Duncan's multiple range test.

*Tissue PPAR-α, PPAR-γ mRNA expression*

We tested whether herbs up-regulate PPAR-α and PPAR-γ in liver and adipose tissues, respectively. Results indicated that liver PPAR-α mRNA expressions were increased by 2.8-fold in mulberry leaf water extract group, 4.0-fold in Korean red ginseng group, 1.9-fold in banaba leaf water extract group, and 5.9-fold in the combination diet group (Fig. 1). PPAR-γ mRNA expressions of adipose tissue were also up-regulated by 37.2-fold in mulberry leaf water extract group, 42.1-fold in Korean red ginseng group, 22.4-fold in banaba leaf water extract group, and 54.2-fold in combination diet group (Fig. 1). These data suggest that increases in both PPAR-α and PPAR-γ mRNA expressions may be related to the regulation of hyperglycemia and dyslipidemia in these animals.

*Tissue LPL mRNA expression*

LPL is a key enzyme in lipoprotein and adipocyte metabolism. As shown in Fig. 1, adipose tissue LPL mRNA expression was increased by the experimental diets containing herbs and the combination diet was most effective. LPL mRNA expressions were increased by 5.1-fold in mulberry leaf water

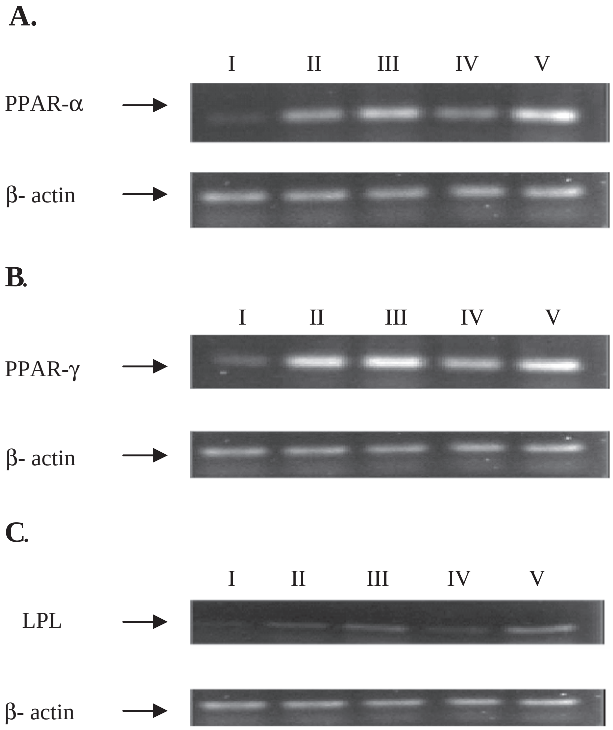


Fig. 1. Tissue PPAR-α, PPAR-γ mRNA expressions of C57BL/KsJ-db/db mice. (A) RT-PCR analysis of PPAR-α mRNA expression in the liver of C57BL/KsJ-db/db mice. (B) RT-PCR analysis of PPAR-γ mRNA expression in the adipose tissue of C57BL/KsJ-db/db mice. (C) RT-PCR analysis of LPL mRNA expression in the adipose tissue of C57BL/KsJ-db/db mice. (I) The control group, (II) the 0.5% mulberry leaf water extract diet group, (III) the 0.5% Korean red ginseng diet group, (IV) the 0.5% banaba leaf water extract diet group and (V) the 0.5% combination diet group.

extract group, 9.3-folds in Korean red ginseng group, 4.2-fold in banaba leaf water extract group, and 20.0-folds in the combination diet group (Fig. 1).

## Discussion

Insulin resistance and obesity are important predictors of NIDDM development. Epidemiological studies have suggested that insulin resistance are associated with cardiovascular disease (CVD) risk factors including elevated triglyceride and decreased HDL cholesterol levels (D'Agostino et al., 2004). Haffner and Cassells (2003) reported that subjects with NIDDM without prior myocardial infarction (MI) have a high risk for CVD as subjects without diabetes with prior MI. Therefore, the management of hyperinsulinemia and dyslipidemia is critical to prevent NIDDM and related CVD complications.

Our study results indicated that mulberry leaf water extract, Korean red ginseng, banaba leaf water extract, and the combination of above herbs effectively control hyperglycemia and hyperinsulinemia by significantly reducing fasting blood glucose and insulin levels in db/db mice. Study results showed that herbs exerted their effects similar to conventional insulin sensitizers, which act as PPAR-α and PPAR-γ agonists. Also, the level of blood glycosylate $HbA1_C$, which is a marker of long-term control of blood glucose, is significantly decreased. Kimura et al. (1999) observed hypolipidemic effects of ginseng water extract in KK-CA(y) obese mouse models (Kimura et al., 1999). Others also have shown that ginseng extract decreases the level of blood lipids (Cicero et al., 2003; Kim et al., 2003; Yokozawa et al., 2004). Inoue et al. (1999) reported that ginseng root saponins sustained LPL activity at a normal level, resulting in the decreases of serum TG and cholesterol (Inoue et al., 1999). *P. ginseng* berry extract exerted antihyperglycemic and anti-obese effects by increasing energy expenditure (Attele et al., 2002). Although a limited number studies are reported for the anti-diabetic effects of mulberry and banaba, recent reports by Andallu et al. (2001), Andallu and Varadacharyulu (2002), and Andallu and Varadacharyulu (2003) showed mulberry feeding is effective in lowering blood lipids and blood glucose concentrations. Liu et al. (2001) showed that banaba extract increases the glucose uptake of 3T3-L1 adipocytes and its possible mechanism is inhibition of adipocyte differentiation.

The present study proved that mulberry, Korean red ginseng, and banaba, traditionally used anti-diabetic herbs, are effective to delay the progress of the disease in a NIDDM model possibly by improving lipid metabolism through the up-regulation of PPARs expressions. PPARs are ligand-activated transcription factors belonging to the nuclear receptor superfamily, and PPAR ligands include fatty acids and eicosanoids (Verges, 2004). PPAR-α agonists are known to stimulate mitochondrial oxidation and cellular uptake of free fatty acids by modifying the expression of genes such as acyl-CoA synthetase gene and fatty acid transport protein gene (Schoonjans et al., 1995; Reddy and Hashimoto, 2001). Ersten et al. (1999) reported that, in a fasting state, cellular uptake of fatty acids liberated from fat tissues occurs with increased liver PPAR-α expression (Ersten et al., 1999). Pharmacological stimulation with synthetic PPAR-α ligands such as fibrates also up-regulates genes involved in fatty acid oxidation and cellular uptake of free fatty acids (Chou et al., 2002; Kim et al., 2003; Sugden and Holness, 2004). PPAR-α ligands also increase the expression of the lipoprotein lipase gene and reduce the expression of the gene encoding for apo-C-III, which is an inhibitor of lipoprotein lipase (Staels et al., 1998) resulting in hypotriglyceridemic effect. One previous report suggested that Korean red ginseng improves serum lipid profiles by stimulating peroximal fatty acid β-oxidation through PPARα-mediated pathways (Yoon et al., 2003).

PPAR-γ has high expression in adipocytes and plays an important role in liver lipid storage and differentiation of fat cells (Tontonoz et al., 1994). In NIDDM patients, TZD, a synthetic PPAR-γ ligand, significantly increased insulin sensitivity (Sood et al., 2000). Lee et al. (2003) have suggested that PPAR-γ ligand up-regulated the expression of genes involved in glucose uptake and lipid storage of adipocytes as well as lipid uptake and storage of the liver (Lee et al., 2003). Therefore, we assumed that the herbs used in our study behave similar to several PPARs ligands. Also, the observed up-regulation of adipocyte lipoprotein lipase and a significant decrease in blood triglyceride by dietary mulberry leaf water extract, Korean red ginseng, banaba leaf extract, or the combination of these herbs are possibly mediated by increased expression of PPARs.

A significant decrease in glycosylated hemoglobin indicates that dietary mulberry, ginseng, and banaba have a long-term control over hyperglycemia. Since we observed decreases in HbA1c prior to decreases in fasting blood glucose, it is possible to assume that these herbs effectively reduced the formation of HbA1c, not only by their blood glucose-lowering effects, but other indirect mechanisms. Previous studies showed that antioxidants such as α-tocopherol and glutathione reduce the level of HbA1c by their antioxidative capacity, not by reducing blood glucose level (Ortwerth and Olesen, 1998; Ceriello et al., 1988).

Our study also suggests that animals fed the herb combination showed the most significant improvement in all of the biomarkers measured. Further studies are required to understand this synergetic effect of these herbs.

## Conclusion

Mulberry leaf water extract, Korean red ginseng, banaba leaf water extract, or their combination fed at a level of 0.5% in the diet significantly increased insulin sensitivity and improved hyperglycemia. Since we used a NIDDM model and test materials were fed as a part of diet, it is possible to suggest that the herbs in the form of dietary supplements possibly improve metabolic syndrome and suppress the development of NIDDM. As far as we know, this is one of very few studies to show that mulberry leaf water extract, Korean red ginseng, and banaba leaf water extract exert their effects through PPARs expression. Future studies are required to investigate responsible compounds in these herbs and possible genes regulated by these PPAR agonists.

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